PHASE 4 OF 8

Project Planning

Stages, milestones, dependencies and risks

Project: LegalEase – AI-Powered Legal Document Generator

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4.1 Purpose of this Phase

The Project Planning phase organizes the work so that LegalEase can be delivered in a logical order. It breaks the project into stages, lists the activities and outputs of each stage, identifies dependencies and risks, and sets out the environment needed by the team.

4.2 Project Objectives

- Deliver a working application that generates, previews, edits and exports legal document drafts.
- Integrate Streamlit, FastAPI, Gemini and the formatting modules into one workflow.
- Verify the complete workflow through testing and present it in a final demonstration.

4.3 Planned Stages

The project was planned in the following stages. Each stage produces a defined output that the next stage depends on.

#	Stage	Main activities	Output
1	Research and model selection	Study generative AI options; select Gemini; review libraries	Technology decision
2	Architecture design	Define components, data flow and API contract	Design specification
3	Backend API development	Set up FastAPI, /generate endpoint, Pydantic validation, Gemini connection	Working API
4	Frontend development	Build Streamlit form, preview and edit area	Working interface
5	Document formatting	Implement TXT, DOCX and PDF generators	Export modules
6	Testing	Test each module and the full workflow	Test results
7	Local deployment	Configure environment and run backend and frontend locally	Runnable system
8	Final demonstration	Present the end-to-end workflow	Demo and documentation

4.4 Work Breakdown

Work is grouped into workstreams so that the four team members can progress in parallel where dependencies allow.

Workstream	Scope	Depends on
Backend and AI	FastAPI setup, request model, Gemini integration, prompt design	Design specification
Frontend	Streamlit layout, input controls, preview, edit, download buttons	API contract
Document formatting	TXT, DOCX and PDF generation, logo and footer	Format of generated text
Testing and documentation	Test cases, defect tracking, setup guide, project documents	Working modules

4.5 Environment and Dependency Plan

The plan includes dependency installation, environment configuration and verification of end-to-end functionality. The following preparation steps are required before development starts:

1. Install Python and create an isolated virtual environment.
2. Install the required packages: Streamlit, FastAPI, Uvicorn, Google Gemini SDK, python-docx, FPDF, Pillow, Requests and python-dotenv.
3. Obtain a Gemini API key and store it in an environment (.env) file.
4. Verify that the backend starts with Uvicorn and the frontend starts with Streamlit.
5. Run a small test request to confirm the connection to Gemini.

4.6 Milestones

Milestone	Completion indicator
M1 – Design approved	Architecture and API contract agreed by the team
M2 – Backend ready	/generate returns generated text for a valid request
M3 – Frontend ready	Form, preview and edit work against the backend
M4 – Exports ready	TXT, DOCX and PDF files download correctly
M5 – Testing complete	All planned test cases executed and defects resolved
M6 – Demonstration	Live end-to-end walkthrough delivered

4.7 Risk Management

Risk	Impact	Mitigation
Gemini API unavailable or key missing	Generation fails	Clear error messages; verify key early in setup
Poor quality of generated text	Draft needs heavy editing	Refine the prompt; provide editable preview; include review notice
Library or version conflicts	Delays in setup	Use a virtual environment and a requirements list
Formatting differences between DOCX and PDF	Inconsistent look	Test both formats with the same sample input

4.8 Quality and Communication Plan

Each module is tested as soon as it is built, and the integrated workflow is verified before the demonstration. The team reviews progress at the end of every stage, records issues found, and confirms that the stage output meets its completion indicator before moving on.

4.9 Roles and Responsibilities

The four-member team shares ownership of the project. The roles below describe the areas of responsibility; members support each other across areas whenever a stage needs extra effort.

Area	Responsibility
Backend and AI integration	FastAPI endpoint, validation, prompt and Gemini connection
Frontend	Streamlit interface, preview, editing and download controls
Document formatting	TXT, DOCX and PDF generators, logo and footer
Testing and documentation	Test cases, defect tracking, setup guide and final documents

4.10 Deliverables

- Working LegalEase application (backend, frontend and formatting modules)
- Requirement, design and planning documents
- Test cases and test results
- Setup instructions and project documentation
- Final demonstration

4.11 Completion Criteria for Each Stage

Stage	Considered complete when
Research and model selection	Gemini and supporting libraries are confirmed as suitable
Architecture design	Components, data flow and API contract are agreed
Backend development	/generate validates input and returns generated text
Frontend development	Users can enter data, see and edit the preview
Document formatting	All three formats download and open correctly
Testing	All planned tests are run and issues fixed
Local deployment	Backend and frontend run together from a clean setup
Demonstration	Full workflow is shown without manual workarounds

4.12 Change Management

If a new idea arises during development, the team compares it with the agreed scope and priorities. Changes that support the core workflow are accepted; extras such as accounts or e-signatures are recorded for future work so that the schedule of the main deliverables is protected.